Power BI Assignment 2 – DAX & Data Visualization

Calculated Columns:

● ***Create a Calculated Column for 'Category Type':*** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

```
Category_Type = CONCATENATE('Order Details (1)'[Category],"-"&'Order Details (1)'[Sub-Category])
```

● ***Calculate Revenue per Order in Order Details Table:*** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

```
Revenue = 'Order Details (1)'[Amount] * 'Order Details (1)'[Quantity]
```

● ***Create a Calculated Column to Categorize Sales:*** Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

```
Sales_Category = IF('Order Details (1)'[Revenue]> AVERAGE('Order Details (1)'[Revenue]),"Above Average","Below Average")
```

Calculated Measures:

● ***Calculate Order Count:*** Define a measure to count the total number of orders in the Order Details table.

```
Count_Orders = COUNTA('List of Orders (1)'[Order ID])
```

● ***Calculate Average Profit in Delhi:*** Create a measure to calculate the average profit for orders placed in Delhi.

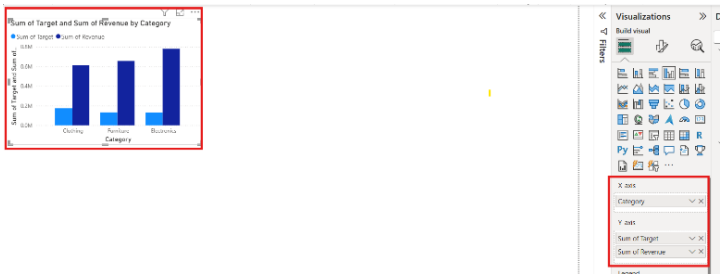
```
Avg_Profit_Delhi = CALCULATE(AVERAGE('Order Details (1)'[Profit]),'List of Orders (1)'[City]="Delhi")
```

● ***Calculate Year-to-Date (YTD) Sales***: Define a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.

```
Year_To_Date = TOTALYTD(SUM('Order Details (1)'[Revenue]),'List of Orders (1)'[Order Date])
```

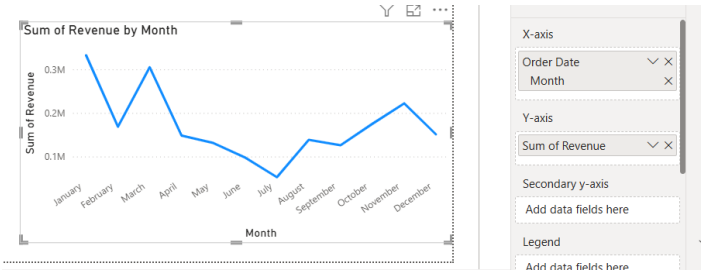
Data Visualization:

Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.

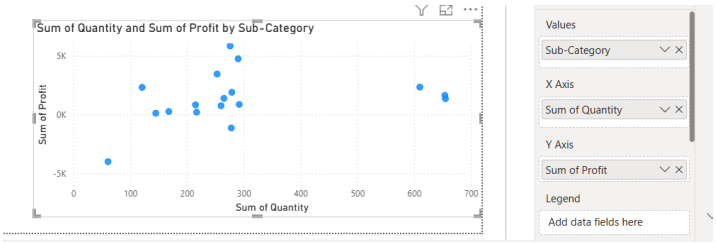


Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub category of products using a donut chart.

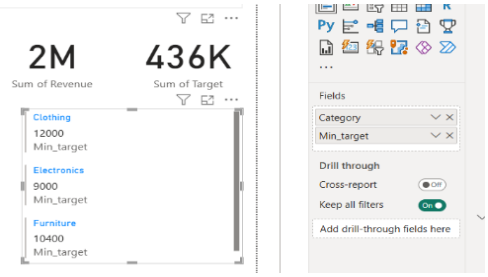
Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



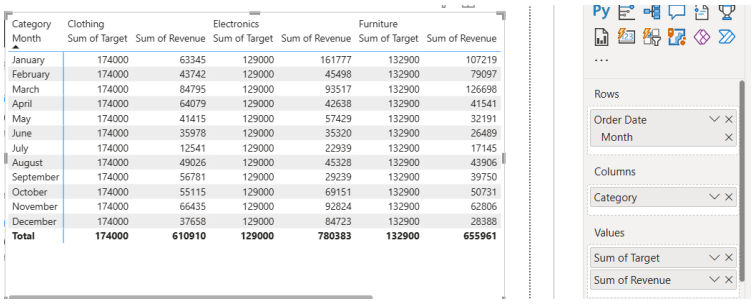
Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

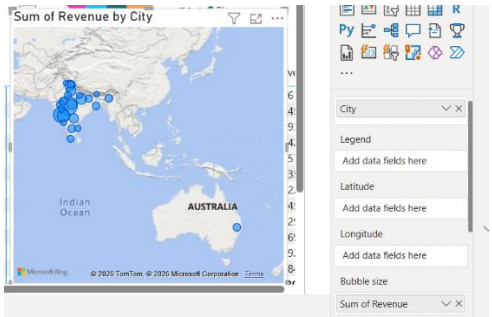


Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

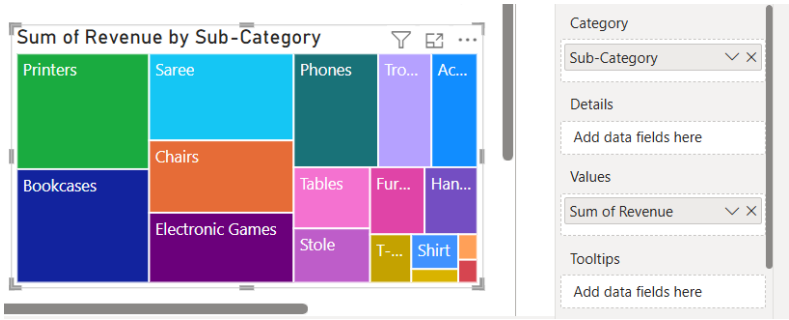
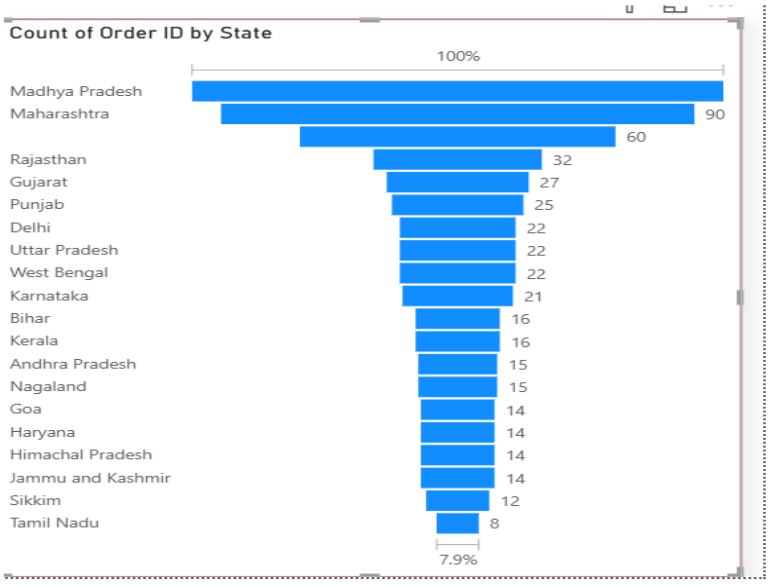
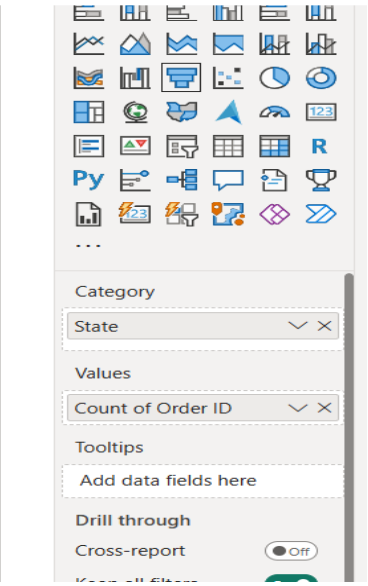


Category	Clothing		Electronics		Furniture	
Month	Sum of Target	Sum of Revenue	Sum of Target	Sum of Revenue	Sum of Target	Sum of Revenue
January	174000	63345	129000	161777	132900	107219
February	174000	43742	129000	45498	132900	79097
March	174000	84795	129000	93517	132900	126698
April	174000	64079	129000	42638	132900	41541
May	174000	41415	129000	57429	132900	32191
June	174000	35978	129000	35320	132900	26489
July	174000	12541	129000	22939	132900	17145
August	174000	49026	129000	45328	132900	43906
September	174000	56781	129000	29239	132900	39750
October	174000	55115	129000	69151	132900	50731
November	174000	66435	129000	92824	132900	62806
December	174000	37658	129000	84723	132900	28388
Total	**174000**	**610910**	**129000**	**780383**	**132900**	**655961**

Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a Treemap.



Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.